

Teorema 1. *Let T be a consistent L -theory and let $\equiv_T$ be the equivalence relation on $\text{Mod}(T)$ given by elementary equivalence. Suppose $\equiv_T$ has exactly n classes for some $n < \omega$. Let M_i with $i = 1, \dots, n$ be representant of each class. Then there exist $\psi_1, \dots, \psi_n$ L -sentences such that $\forall i = 1, \dots, n \ T \cup \{\psi_i\} \vdash \text{Th}(M_i)$.*

Proof. Let $(\phi_{i,j})_{1 \leq i < j \leq n}$ be L -sentences such that $M_i \models \phi_{i,j}$ and $M_j \not\models \phi_{i,j}$. Now consider $\psi_i := \bigwedge_{k < i} \neg \phi_{k,i} \wedge \bigwedge_{k > i} \phi_{i,k}$. Then $T \cup \{\psi_i\} \vdash \text{Th}(M_i)$. □